VISHNU PURUSHOTHAMAN

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EDUCATION

The University of Texas at Austin Expected May 2029
Bachelor of Science in Computer Science | GPA: 3.93 / 4.00 *Austin, TX*
Relevant Coursework: Data Structures, Discrete Mathematics, Computer Organization & Architecture, Linear Algebra, Statistics & Data Science, Multivariable Calculus

PROJECTS

SnapNotes — iOS piano-learning app (published) | App Store | Website Summer 2026 – Present
Flutter/Dart, Swift, Python/FastAPI, Docker, Fly.io, Audiveris OMR, StoreKit, CoreMIDI, AVFoundation

- Designed, built, and shipped a solo iOS/iPadOS app (1k+ downloads) that converts photographed or PDF sheet music into an interactive falling-notes practice experience; monetized with a free tier and Pro subscription via StoreKit
- Architected a Flutter front end (BLoC state management, custom-drawn 60 fps note visualizer, MIDI keyboard input over USB/Bluetooth) with native Swift integrations for document scanning, home-screen widgets, and iCloud backup
- Deployed an optical music recognition backend (FastAPI, Audiveris, Tesseract OCR) in Docker on Fly.io with token auth, scale-to-zero, and result caching; maintained a 254-song catalog and ~370 automated tests through TestFlight and App Store release

Moontide — iOS/iPadOS arcade game (published) | App Store | Website Sep 2026
Swift, SwiftUI, Metal, CoreHaptics, AVAudioEngine, StoreKit 2, Game Center, iCloud, XCTest

- Built a one-touch physics game solo, prototyped on the web then ported to native Swift with zero dependencies; golden-trace tests replay 1,100 recorded frames to verify parity with the original to within 1e-11 px
- Wrote a custom Metal water shader and velocity-driven CoreHaptics on a 120 Hz render loop; split the game into Core/Rendering/Audio/Platform packages with an event-driven core covered by 130+ tests; shipped 4 updates in 2 weeks

Companion: Diabetes Journal — iOS + watchOS health app (published) | App Store Oct 2026
Swift 6, SwiftUI, SwiftData, HealthKit, CloudKit, Foundation Models, StoreKit 2, WidgetKit, App Intents

- Built a local-first diabetes journal solo under Swift 6 strict concurrency: iPhone app, Apple Watch app with complications, Lock Screen widgets, Live Activities, Control Center controls, Siri App Intents, and CloudKit sync
- Shipped an on-device conversational assistant on Apple's Foundation Models framework with a tool-grounded session (10 data tools), streaming replies, and deterministic routing gates so the model can only cite real logged data
- Designed a deterministic insight engine and glucose-analytics layer (ambulatory glucose profile, variability, food response) verified by a 127-test Swift Testing suite; owned StoreKit 2 subscriptions and a CLI archive-and-upload pipeline (16 builds)

Agentville — HackTX 2026, AI agents that build live web apps | Live demo | GitHub Oct 2026
TypeScript, React Three Fiber, Fastify, WebSockets, Claude Agent SDK, TimescaleDB, zod, Vultr

- Built a 3D voxel world where placing and wiring buildings produces a real, deployed web app, coded live by sandboxed Claude agents (Haiku triage, Sonnet coding); every on-screen animation is driven by a real tool call
- Added self-healing deploys (agent fixes from real stack traces, server-side test reruns, blue/green redeploys with zero dropped Nessie payments); load-tested to 500 visitors and 19,819 requests with 0 errors and a 3 ms median

Talk Up — speech-support iOS app 2025

- Co-developed a Swift/UIKit app using CoreML to give users with speech disabilities real-time feedback on clarity, tone, and pacing

EXPERIENCE

Android App Development Intern — The University of Texas at Dallas, Dept. of Computer Science Jun – Aug 2024

- Developed multiple Android apps in Java/XML, culminating in a multithreaded game with an enhanced Pong ruleset; recognized by Prof. John Cole for rapid mastery of advanced concepts

CS Entrepreneurship Intern — Science Gurus (Remote) Jun – Jul 2024

- Completed a 6-week intensive on product strategy, market research, UX design, prototyping, and go-to-market planning

LEADERSHIP & ACTIVITIES

Texas Game Jam — Participant Sep 2025, Sep 2026

- Built a solo 2D game in Unity/C# in 48 hours at a statewide event judged by industry professionals
- 2026: co-built Return to Sender in Godot, a parry-based bullet-hell with 3 multi-phase bosses, 4 modes, and 39 voiced lines

Flower Mound HS Computer Science Club — Officer 2022 – 2025

- Organized hackathons and mentored peers; placed 6th (team) in the HPE CodeWars Novice Division

Flower Mound HS Symphonic Orchestra — Concertmaster Violinist 2021 – 2025

- Led a 50+ member orchestra as concertmaster, running rehearsals and coordinating section leaders

SKILLS

Languages: Swift, Dart, Java, Python, TypeScript, JavaScript/HTML/CSS, C#, SQL
Mobile & Frameworks: SwiftUI, UIKit, SwiftData, HealthKit, CloudKit, WidgetKit, App Intents, StoreKit 2, CoreML, CoreMIDI, AVFoundation, Metal, CoreHaptics, Foundation Models, Flutter (BLoC), Android (Java/XML), Unity, Godot, FastAPI
Tools & Infrastructure: Git, Docker, Fly.io, Xcode, App Store Connect/TestFlight, Swift Testing, XCTest, Audiveris, Tesseract, music21
Certifications: Python for Everybody (University of Michigan); Mathematics for Machine Learning (Imperial College London)